

Final Project: Draft Report

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Introduction

Baseball is a game of inches, and we all know that the small details of the game do matter to make optimal roster decisions that go in favor of your organization and benefit the overall baseball game. We have experienced the development of different baseball eras in which we have had inning eaters, flame throwers, knuckleballers, and have even seen the extinction of pitches like the eephus pitch. Not to mention the technological advancements from the implementation of pitching biomechanics to the evolution of pitching metrics during the Statcast era since 2015. Indeed, baseball has evolved in an escalated manner and has undergone many regulations and rule changes that indeed affect the pitching workload, such as the implementation of the Pitch Clock and ABS system.

The Pitch Clock and ABS system do indeed increase the number of pitches thrown by a pitcher and how fast he needs to recover between one pitch and the next. Even though both measures highly impact the industry's marketability by reducing the time of baseball games and improving umpire calls, they do increase the possibility of a pitcher's IL placement. However, we also know based on MLB's league reports that the rise in pitcher injuries is also driven by the chase for more velocity, nastier stuff (via pitch design), and max-effort pitching (Adler, 2024). Hence, it was essential for us to take some time to unravel additional hidden insights, trends, and statistical significance of these inferences through the effective incorporation of the IL placement dataset containing data from 2015 to 2017, to project how and if these factors truly play a role in the detrimental effects on a pitcher's health.

Upon the appropriate conduction of EDA analysis through the initial report last week, we developed a robust structure that served as a foundation to proceed with an adequate analytical framework that could be further used by teams trying to mitigate pitching injuries and stay away from high-risk profiles. It could also serve as a guide for teams in their means of limiting pitcher outings and shutting down highly coveted prospects to protect their arms as much as possible if this given pitcher is prone to high-risk injuries. Complementing the results of the initial analysis, which suggested that detrimental factors also affecting a pitcher's longevity in today's game may include elevated walk rates and potentially excessive workloads. At the same time, strikeout ability and controlled velocity appear to be beneficial to effectiveness; combined with this week's proven statistical procedures, they raise the significance of our analysis in many ways. Knowing that while velocity remains a performance asset, balancing it with control and workload management is a critical aspect taken into consideration for the data complementation of IL placement.

Data Preparation and Integration

Upon the completion of the data preparation stage, which was thoroughly explained throughout the introduction of the initial analysis, we proceeded to integrate the IL data for this week's deliverable in which it was deemed necessary to create various data subsets in means of properly preparing the data to be merged and adequately cleaning attribute naming structure, eliminating invalid and erroneous data and building join key helpers to integrate data across frames efficiently. Once this initial cleansing stage was demonstrated to be ready for further debugging, we produced an additional cleaning subset in which it was necessary to limit the data to the relevant years (2015-2017), based on MLB's publicly available dataset regarding IL placement. Regardless of knowing that this data included position players in it, we didn't focus

on eliminating them, as they were going to be evidently limited when merging the IL dataset to the already set `pitcher_df` containing pitchers with at least 2000 pitches obtained via the Savant dataset incorporation.

Before merging this IL placement data with what we have already constructed, it was essential to add relevant variables to it, built from its already included original data. This is why, at this point, we created a final subset to include only the relevant data for merging considerations. Here, we used `join_key_fl` as the standardized name variable and included an `injury_count` variable to reflect the amount of injuries this given pitcher had in this three seasons timeframe; `total_dl_days` which summarized the amount of days this given pitcher spent on the IL list; `last_injury_dt` to showcase the last date of this player's injury placement and `injury_types` to note the types of injuries the player has surpassed. Upon completion of this stage, it was necessary to once again check dupes in our data frame, and it was confirmed that no duplicates were part of this IL data, which enabled it to be ready for merging purposes.

To properly merge both data frames, we mutated the names in `pitcher_df` to standardize its naming structure and make it simulate the same structure as that of the IL names, which displayed 'First Last names' instead of 'Last, First' as the `pitcher_df` was initially structured. Then, through a left join, we merged the data. Instead of including the types of injuries and the last date of injury, we only included the number of injuries; the total days spent in the injured list; and the injury occurrence identified as the binary variable to be further used in the logistic regression analysis. Ultimately, we dropped the naming helpers and produced a final clean data structure ready to conduct the logistic regression analysis detailed below.

Methods

Besides the methods of an EDA and the two multivariable linear regressions in the Initial Analysis Report, two logistic regression models were developed to predict injury occurrence using workload and performance predictors. This binary model predicts the probability of a pitcher sustaining an injury by analyzing workload intensity and pitching characteristics. Additional diagnostics such as multicollinearity tests, confusion matrices, ROC Curve, and AUC were used to ensure model validity and interpretability.

Analysis

Splitting the Data

The dataset was split into a training and a test set. The training dataset consists of the data on which the model will be trained, and the test set will be used to evaluate the model. This helps ensure that the model performs effectively on new data and that we're not overfitting the model. Therefore, 70% of the observations are used for the training set, and 30% are used for testing, ensuring the injury occurred response variable remains balanced across both sets. The training set comprised 578 observations with 48 variables, while the test set contained 248 observations with 48 variables.

Logistic Regression Models

Two logistic regression models were fitted to the training data. The first logistic regression model included the variables: total_pitches, velocity, spin_rate, IP_total, SO9_career, and BB9_career. While the second logistic regression model had the same variables, except that IP_total was removed, we will explain why that is. These logistic regression models predict the

probability that a pitcher experienced an injury using key metrics about their workload and performance.

Logistic Regression Model 1: Predicting Injury Occurrence

The first logistic regression model was built to predict injury occurrence, which included variables: total_pitches, velocity, spin_rate, IP_total, SO9_career, and BB9_career.

Figure 1

```
Call:
glm(formula = injury_occurred ~ total_pitches + velocity + spin_rate +
    IP_total + SO9_career + BB9_career, family = binomial, data = train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.2979164  3.1990968   1.031   0.303
total_pitches  0.0007072  0.0001324   5.341 9.25e-08 ***
velocity     -0.0334674  0.0376229  -0.890   0.374
spin_rate    -0.0002222  0.0005472  -0.406   0.685
IP_total     -0.0090854  0.0019997  -4.543 5.54e-06 ***
SO9_career   -0.0706486  0.0614210  -1.150   0.250
BB9_career   -0.0842542  0.1241712  -0.679   0.497
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 784.58  on 577  degrees of freedom
Residual deviance: 714.40  on 571  degrees of freedom
AIC: 728.4

Number of Fisher Scoring iterations: 4
```

Most variables were not statistically significant; only variables total_pitches and IP_total were statistically significant. The p-values (9.25e-08) and (5.54e-06) were less than 0.01. So, as total pitches increased, the pitcher's chance of getting injured increased. This was positive and significant, meaning the more pitches a pitcher has, the higher the chance of them getting injured is. However, IP_total was negatively significant, suggesting that pitchers with more career innings pitched might be less prone to injury, which could possibly indicate better conditioning or durability. The AIC (728.4) was notably high as well. This suggested that something was wrong with the model.

So, before fitting the final logistic regression model, multicollinearity was assessed, and the VIF values for the predictors were:

total_pitches	velocity	spin_rate	IP_total	SO9_career	BB9_career
42.278779	1.248559	1.149305	43.837395	1.294158	1.403822

Both total_pitches and IP_total showed high VIF values, which indicated strong multicollinearity, as both are metrics that capture pitcher workload. To improve model stability and interpretability, only total_pitches was retained in the final model, since it offers a direct measure of cumulative workload and injury risk.

Logistic Regression Final Model: Predicting Injury Occurrence

The final logistic regression model was fit to predict injury occurrence among pitchers using total_pitches, velocity, spin_rate, SO9_career, and BB9_career as the predictors.

Multicollinearity was also checked for this model:

total_pitches	velocity	spin_rate	SO9_career	BB9_career
1.121891	1.220109	1.143447	1.295314	1.238973

Since the VIF values are less than 5, this suggests that multicollinearity isn't a problem for this model. The independent variables in the regression model are not correlated.

Figure 2

```
Call:
glm(formula = injury_occurred ~ total_pitches + velocity + spin_rate +
     SO9_career + BB9_career, family = binomial, data = train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  5.350e+00  3.143e+00   1.702   0.0887 .
total_pitches 1.216e-04  2.117e-05   5.743 9.29e-09 ***
velocity     -6.184e-02  3.677e-02  -1.682   0.0926 .
spin_rate    -3.145e-04  5.346e-04  -0.588   0.5563
SO9_career   -8.324e-02  6.108e-02  -1.363   0.1729
BB9_career    1.322e-01  1.140e-01   1.159   0.2463
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 784.58  on 577  degrees of freedom
Residual deviance: 737.34  on 572  degrees of freedom
AIC: 749.34

Number of Fisher Scoring iterations: 4
```

The model revealed that total pitches were again highly statistically significant and had a lower p-value ($9.29\text{e-}09$) than the previous model. Velocity in this model showed a marginally negative association with injury occurrence. However, the rest of the predictors again were not statistically significant. The AIC (749.34) is a little higher in this model; however, even though the AIC was a little higher than the first model, this model improves interpretability and is more stable, as there is no multicollinearity present.

Since the regression coefficients are in log-odds, we decided to convert them to odds for interpretability. But before we did, these were our log-odd results:

(Intercept)	total_pitches	velocity	spin_rate	SO9_career	BB9_career
5.3498762786	0.0001215989	-0.0618442511	-0.0003145439	-0.0832426375	0.1321859697

and these were our odds results:

(Intercept)	total_pitches	velocity	spin_rate	SO9_career	BB9_career
210.5822427	1.0001216	0.9400293	0.9996855	0.9201279	1.1413206

The intercept is 210.5822427, meaning that when all the predictors were zero, the injury odds for a pitcher are 210.5822427. For total_pitches (1.0001216), it means that for each additional pitch, the odds of injury increase by a factor of 1.0001216, while holding other variables constant. For velocity (0.9400293), it means that for each additional 1 mph increase in average velocity, the odds of injury decrease by a factor of 0.9400293, while holding other variables constant. Though this is marginally significant, it may have a slight effect on injury odds. For spin_rate, SO9_career, and BB9_career, these essentially were not significant, so they didn't predict a pitcher's injury odds. It seems that total_pitches were the strongest and most reliable predictor of injury, which means workload was the dominant predictor of injury risk. To further illustrate the model's predictions, probabilities were calculated by varying one predictor while holding others at their mean value. In this case, since total_pitches is highly statistically significant, probability changes for different values of total pitches were observed.

Figure 3

	total_pitches	velocity	spin_rate	S09_career	BB9_career	probs
1	2001.000	88.89528	2223.019	8.549378	3.31555	0.2937776
2	6265.996	88.89528	2223.019	8.549378	3.31555	0.4113256
3	25820.000	88.89528	2223.019	8.549378	3.31555	0.8828041

So, as the value of total_pitches increased from its minimum (2001) to maximum (25820), the probability of a pitcher getting injured increased from (0.2937776) to (0.8828041). These results clearly show that cumulative workload is strongly associated with a higher risk of injury, even when considering other performance-related metrics such as velocity, spin rate, strikeout rate, and walk rate.

Confusion Matrix and Model Performance: Training Set

Next, we created a confusion matrix, and the logistic regression model was assessed on the training set with a 0.5 classification threshold. The confusion matrix showed that there were 283 true negatives (TN), so predicted no injury, and the pitcher did not get injured, 158 were false negatives (FN), so predicted no injury but the pitcher did get injured, 55 were false positives (FP), so predicted injury but the pitcher did not get injured, and 82 were true positives (TP), so predicted injury and the pitcher did get injured. So, the model correctly identified 283 pitchers who did not get injured and 82 pitchers who did get injured, and the misclassifications seemed to be imbalanced, between 158 false negatives and 55 false positives. The confusion matrix indicated that the logistic regression model on the training set achieved an accuracy of 63%, so the model correctly predicted injury status 63% of the time. A precision of about 60%, so when the model predicted an injury, it was right about 60% of the time. A recall of 34%, suggesting that the model correctly identified roughly one-third of pitchers who sustained injuries. Specificity was higher at about 84%, indicating that the model was better at correctly identifying healthy players. While the model demonstrates moderate accuracy, its ability to

identify injured players could be improved. Nonetheless, the significant contribution of total pitches as a predictor supports the conclusion that workload significantly increases injury risk for pitchers.

Figure 4

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	283	158
Yes	55	82

Accuracy : 0.6315
 95% CI : (0.5907, 0.6709)
 No Information Rate : 0.5848
 P-Value [Acc > NIR] : 0.01229

 Kappa : 0.1908

 McNemar's Test P-Value : 2.77e-12

 Sensitivity : 0.3417
 Specificity : 0.8373
 Pos Pred Value : 0.5985
 Neg Pred Value : 0.6417
 Prevalence : 0.4152
 Detection Rate : 0.1419
 Detection Prevalence : 0.2370
 Balanced Accuracy : 0.5895

 'Positive' Class : Yes

Confusion Matrix and Model Performance: Test Set

Now it was time to create another confusion matrix, but this time the logistic regression model was assessed on the test set. The confusion matrix on the test set revealed that there were 138 true negatives (TN), so predicted no injury, and the pitcher did not get injured, 57 were false negatives (FN), so predicted no injury but the pitcher did get injured, 18 were false positives (FP), so predicted injury but the pitcher did not get injured, and 35 were true positives (TP), so predicted injury and the pitcher did get injured. On the test set, the model achieved an accuracy of 70%, slightly higher than the 63% observed in the training data. A precision of 66%, a recall of 38%, and a specificity of about 88%.

Figure 5

Confusion Matrix and Statistics

```

Reference
Prediction No Yes
No 138 57
Yes 18 35

Accuracy : 0.6976
95% CI : (0.6363, 0.7541)
No Information Rate : 0.629
P-Value [Acc > NIR] : 0.01413

Kappa : 0.2903

McNemar's Test P-Value : 1.145e-05

Sensitivity : 0.3804
Specificity : 0.8846
Pos Pred Value : 0.6604
Neg Pred Value : 0.7077
Prevalence : 0.3710
Detection Rate : 0.1411
Detection Prevalence : 0.2137
Balanced Accuracy : 0.6325

'Positive' Class : Yes

```

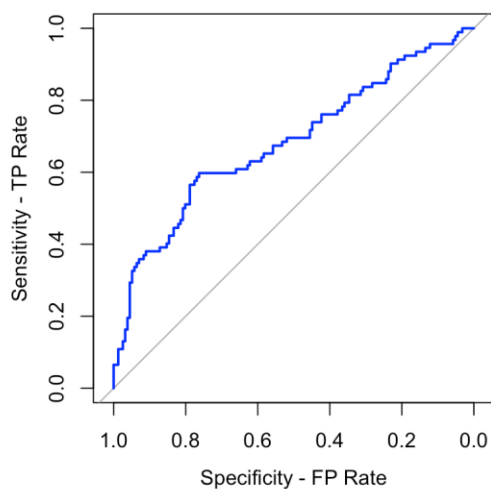
The results compared to the training set are fairly consistent, indicating that the model generalizes well to unseen data (test set) and is not overfit. However, it underperforms while using the train set as the model might be learning the expected outcomes instead of properly running the statistical analysis. These findings again illustrate that the model captures meaningful relationships between workload and the probability of injury; however, improvements in identifying injury cases could be achieved.

ROC Curve and AUC

We then created a ROC Curve for the test set to illustrate the trade-off between sensitivity (true positive rate) and specificity (false positive rate) across different classification thresholds. The ROC curve line lies above the diagonal reference line, suggesting a separation and distinction between injured and non-injured pitchers. Even though it doesn't do it perfectly, logistic regression can meaningfully rank pitchers by injury risk, and it does it better than random guessing. The area under the curve (AUC) was 0.682, suggesting that the model has fair

discriminatory power. This indicates that the logistic regression model can correctly distinguish between injured and non-injured pitchers by approximately 68% of the time. Although the model is not highly discriminative, this performance level is reasonable considering the complex and multifactorial nature of sports injuries.

Figure 6



Conclusion

Conclusively, the information obtained through the effective development of the logistic regression analysis upon the incorporation of IL placement was enticing as it permitted us to clearly visualize those factors truly resulting into an adverse effect towards pitcher's health and the importance of managers, coaches and team executives in having a proper risk management plan in place to prevent the drastic amount of injuries from occurring. It is known that 2025 has been a year marked with injuries due to the increase in "mas-effort"; this is something that goes beyond the scope of the data being analyzed, but definitely correlates with the unresolved results demonstrating that pitching workload is the most crucial factor that eventually all organizations must take into account. Before all these analytical metrics came into place, pitchers were not so interested in outperforming their metrics in each appearance as they are now, which is indeed a

reason why they used to throw more innings and pitches in the past without getting injured.

However, we should also consider that, as indicated by the MLB releases report on injuries to pitchers, the best predictor of future injury is past injury (Adler, 2024). Something that will be further assessed in the final analysis, were we intend to perform additional inferential analysis of the data, including the identification of some names which come tome of the list of high-injury risk and relate it to real-life situations were guys that have been in the injury occurrence list more than once tend to have shortened careers as they are prone to getting injured again shortly after.

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